INCH POUND

MIL-C-3098/83D
5 May 1993

SUPERSEDING
MIL-C-3098/83C
2 November 1983

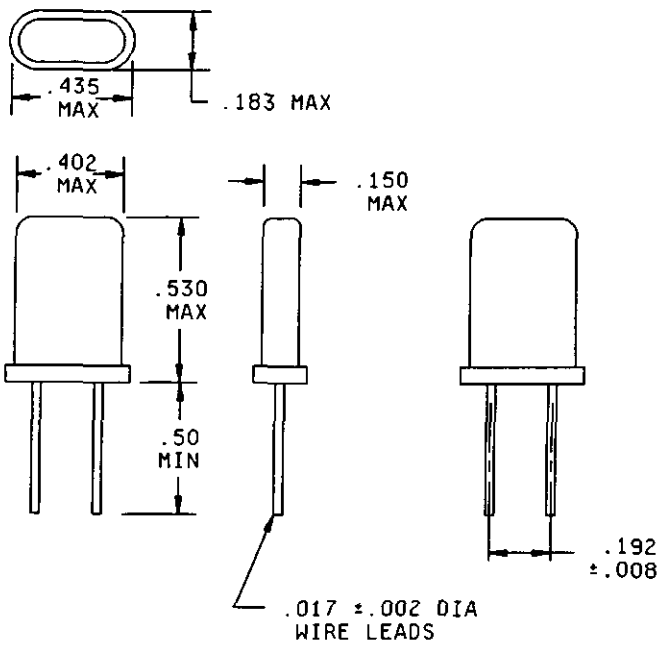
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR107/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 50 to 125 MHz; fifth mechanical overtone; noncontrolled; series resonance.



Inches	mm
.002	.05
.008	.20
.017	.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR107/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-18/U (MIL-H-10056/9) or HC-43/U.

Dimensions, marking, and configuration: See figure 1.

Frequency range: 50 to 125 MHz, inclusive.

Capacitance, shunt: 6.0 ±0.5 pF.

Mode of oscillation: Fifth mechanical overtone.

Rated drive level: 2.0 ±0.4 mW.

Resonance: Series.

Operating temperature range: -55°C to +105°C, inclusive.

Frequency tolerance: ±50 ppm.

Equivalent resistance: 60 ohms, maximum.

Unwanted modes: Method II, the minimum unwanted mode effective resistance shall be at least three times that of the main mode, or 75 ohms, whichever is greater.

Shock: (specified pulse).

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 204 of MIL-STD-202, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

User activities:
Air Force - 17

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0678-17)

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